
Chapter 5: Number Play

Worksheet 5: Divisibility, Number Puzzles, and Algebraic Patterns

Learning Objective: Students will explore divisibility using algebraic generalisation; understand the general forms of numbers (e.g. a 2-digit number = $10a + b$); verify divisibility rules algebraically; and solve number puzzles and magic squares.

Worked Example

A two-digit number has tens digit a and units digit b . The number = $10a + b$.

When digits are reversed: $10b + a$.

Difference = $(10a + b) - (10b + a) = 9(a - b)$, always divisible by 9.

Example: 73 reversed is 37; $73 - 37 = 36 = 9 \times 4$.

Questions 1–5: Easy / Recall

Q1. Write a 2-digit number where tens digit is p and units digit is q in algebraic form.

Q2. A number is written as $100a + 10b + c$. What does this represent?

Q3. Verify: The sum of a 2-digit number and its reverse is always divisible by 11.

Q4. What is the sum of all digits of any multiple of 9? Give two examples.

Q5. A magic square has the property that all rows, columns, and diagonals have the same sum. Verify for:

$$\begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$$

Questions 6–10: Basic Application

- Q6.** The tens digit of a number is 3 more than the units digit. Write the number in terms of the units digit b .
- Q7.** Prove algebraically that the difference of a 3-digit number and its reverse is always divisible by 99.
- Q8.** A number $N = 100a + 10b + c$. Show that N is divisible by 3 if $a + b + c$ is divisible by 3.
- Q9.** Fill in the blank to make a magic 3×3 square using numbers 1–9. (Teacher Verification Required)
- Q10.** Pick any 3-digit number, repeat its digits to make a 6-digit number (e.g. $345 \rightarrow 345345$). Divide by 7, then 11, then 13. What do you notice?

Questions 11–15: Practice and Skill Building

- Q11.** A 2-digit number is 4 times the sum of its digits. If the units digit is 2 more than the tens digit, find the number.
- Q12.** Show that any 3-digit number of the form $\overline{aba}$ is always divisible by 11. (Hint: $\overline{aba} = 100a + 10b + a = 101a + 10b$.)

- Q13.** Using algebra, prove that the product of any two consecutive odd numbers is always odd.
- Q14.** Create a 4-digit number where the first two digits equal the sum of the last two digits. Give an example and verify.
- Q15.** The sum of three consecutive integers is 99. Find them. Now generalise: if three consecutive integers sum to S , what is the middle integer?

Questions 16–18: Higher Order Thinking

- Q16.** Choose any number. Multiply by 3. Add 6. Divide by 3. Subtract original number. You always get 2. Prove this algebraically.
- Q17.** Using n for any integer, prove that $n(n + 1)$ is always even. Hence show $n(n + 1)(n + 2)$ is always divisible by 6.
- Q18.** A number trick: Think of a number. Double it. Add 10. Halve the result. Subtract the original number. The answer is always 5. Prove algebraically.

Questions 19–20: Real-Life Applications

Q19. A bank PIN is a 4-digit number where: the thousands digit is twice the units digit, the hundreds digit equals the tens digit, and the sum of all digits is 18. Find all possible PINs.

Q20. In a cricket match, a batsman's score over 5 overs was 5 consecutive integers. The total was 100. What were the scores per over? Is any such total always a multiple of 5? Prove it.

Reflection Question: Can you create your own number trick that always gives a fixed answer? Write it out and prove why it works using algebra.